



Agate

△ Fire agate with unusually fine colouring

The microcrystalline, compact variety of quartz, agate is a common, semiprecious chalcedony. Agate is mostly characterized by colour bands in a concentric form, and less often by moss-like inclusions, when it is called moss agate. Other names, such as fire, or Brazilian, often precede the word agate and these can describe the locality where they are found, or denote a particular appearance or coloration. Agates are almost always cut *en cabochon*, carved, or used as beads or ornaments.

Specification

Chemical name	Silicon dioxide	Formula	SiO ₂	Colours	All
Structure	Trigonal	Hardness	7	SG	2.6
		RI	1.53–1.54		
Lustre	Vitreous	Streak	White	Locations	Worldwide, notably Brazil, Botswana, South Africa, Egypt, Mexico, China, and Scotland; fire agate only in northern Mexico and southwestern USA

Rough

Agate slice | Seen here in cross-section, the bright and varied circles of colour displayed in this agate slice indicate that it began life as a nodule of layered carnelian, and was then overlain by multiple layers of varicoloured chalcedonies.



Chalcedony layers

Crystalline layers



Brazilian agate | This slice shows stages of the agate's formation: the lava cavity was lined with crystallized quartz, then the inner hollow was layered with varicoloured chalcedony.



Iris agate slice | The chalcedony layers in this agate are extremely thin, and act as diffraction gratings to produce rainbow colours according to their thickness.



Classic bubbly botryoidal surface

Agate rough | The surface of this specimen is botryoidal – it appears as a mass of globular forms, resembling a bunch of grapes. The layering typical of many types of agate is also visible.

Carnelian interlayered with chalcedony